1. Identifying Key Variables and Constraints in Real-World Problems

(Corresponds to Sections 6 and 12 of formulate.pdf)

- Question Content: The problem describes a circular revolving door with a diameter of 2 meters (200 cm), divided into three equal sectors by three wings. It asks students to calculate the maximum allowable arc length of each doorway opening (indicated by dashed arcs in the diagram) to prevent direct airflow between the entrance and exit.
- Relevance Analysis:
  - Students must identify key variables: the circle's diameter (200 cm), number of wings (3), sector angle (120°), and the unknown maximum arc length.
  - They must also recognize the core constraint: if the combined arc lengths of two adjacent openings exceed the arc length of one sector, the entrance and exit become directly connected, allowing uncontrolled airflow. Therefore, the maximum arc length for a single opening cannot exceed half the arc length of one sector.
  - This directly aligns with formulate.pdf's emphasis on "identifying key variables in real-world contexts" and "recognizing constraints, assumptions, and simplifications in mathematical models."

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2. Translating Real-World Situations into Standard Mathematical Representations

(Corresponds to Sections 4, 9, and 10 of formulate.pdf)

- Question Content: Students must extract information from textual descriptions and a top-down diagram to construct a geometric model.
- Relevance Analysis:
  - The task requires students to "decode text and visual representations to build a contextual model" (Section 4).
  - They must abstract the revolving door structure into a geometric figure: a circle divided by three radii into three 120° sectors (Sections 5 and 8).
  - Furthermore, they must translate the physical requirement of "preventing airflow" into a mathematical condition: 2 × arc length ≤ arc length of one sector, or equivalently, 2 × arc length ≤ (120°/360°) × circumference (Sections 9 and 10).
  - This process exemplifies the core skill of "translating problems into standard mathematical representations or algorithms."

3. Simplifying Problems to Enable Mathematical Analysis

(Corresponds to Section 7 of formulate.pdf)

- Question Content: A real revolving door involves complex factors like airflow dynamics, material thickness, and rotational speed, but the problem reduces it to a pure geometry question.
- Relevance Analysis:
  - Students must ignore irrelevant details (e.g., door thickness, wind pressure) and focus solely on spatial partitioning and angular relationships—the essential mathematical structure.
  - This reflects formulate.pdf's principle of "simplifying or decomposing a situation to make it amenable to mathematical analysis," a crucial step in modeling.

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4. Recognizing Mathematical Structures and Patterns

(Corresponds to Sections 5 and 8 of formulate.pdf)

- Question Content: The three wings divide the circle symmetrically into three equal sectors.
- Relevance Analysis:
  - Students must recognize the symmetry of the circle and the regular pattern of equal sector division, leading to the conclusion that each sector subtends a central angle of 120°.
  - This ability to "identify mathematical structures (e.g., patterns, relationships, regularities) within a problem" (Section 5) is foundational to successful modeling.

1. 识别现实问题中的关键变量与约束条件（对应《formulate.pdf》第6、12条）

- 题目内容：题目描述了一个直径为2米（200厘米）的圆形旋转门空间，由三片翼板均分为三个扇区。要求计算每个门开口（图中虚线弧）的最大弧长，以确保入口和出口之间不会因开口过大而形成空气自由流通的通道。
- 相关性分析：
  - 学生必须识别出关键变量：圆的直径（200 cm）、翼板数量（3片）、扇区角度（120°）、以及待求的最大允许弧长。
  - 同时需理解核心约束条件：当两个相邻开口的总弧长超过一个扇区所对应的弧长时，入口与出口就会连通，导致空气流通。因此，单个开口的最大弧长不能超过该扇区弧长的一半。
  - 这直接体现了《formulate.pdf》中"识别现实问题中的关键变量"和"识别数学模型中的约束、假设与简化"的要求。

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2. 将现实情境转化为标准数学表示（对应《formulate.pdf》第4、9、10条）

- 题目内容：学生需要从文字描述和俯视图中提取信息，并将其转化为几何模型。
- 相关性分析：
  - 题目要求学生"读懂并解码文字、图像以构建情境模型"（第4条）。
  - 必须将旋转门结构抽象为一个圆被三条半径等分为三个120°扇形的几何图形（第5、8条）。
  - 进而将"防止空气流通"的物理要求转化为数学条件："两个开口的总弧长 ≤ 一个扇区的弧长"，即 2 × 弧长 ≤ (120°/360°) × 圆周长（第9、10条）。
  - 这一过程完美体现了"将问题翻译为标准数学表示或算法"的能力。

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3. 简化问题以便进行数学分析（对应《formulate.pdf》第7条）

- 题目内容：现实中的旋转门涉及复杂的气流、材料、动力学等因素，但题目将其简化为一个纯几何问题。
- 相关性分析：
  - 学生需忽略无关细节（如门的厚度、转速、风压等），聚焦于空间分割与角度关系这一核心数学结构。
  - 这种"简化或分解情境以使其适合数学分析"的做法，正是《formulate.pdf》第7条所强调的建模关键步骤。

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4. 识别问题中的数学结构与规律（对应《formulate.pdf》第5、8条）

- 题目内容：三片翼板将圆三等分，形成对称结构。
- 相关性分析：
  - 学生需识别出圆的对称性和等分扇形的规律，从而推导出每个扇区对应120°的中心角。
  - 这种对"问题中数学结构（如规律、关系、模式）的识别"（第5条）是成功建模的前提。